LAKSHIMI RAMAN S

Database Administrator / Backend Engineer / Distributed Systems Engineer

☒ lakshimiramans@gmail.com **in** lakshimi-raman-s **GH** LakshimiRam-073 • Chennai, India

SUMMARY

Database Administrator on Zoho's SaaS Framework team (15 engineers), working on PostgreSQL and MySQL infrastructure, automated failover, zero-downtime upgrades, replication tooling, and on-call production support for 50+ products and 50M+ tenants across 900+ clusters and 54 data centers. Strong in Go, Java, C, AWS, system design, and database internals.

EXPERIENCE

Database Administrator, SaaS Framework Team
Zoho Corporation Jun 2025 – Present
Chennai, India

- Administer 900+ PostgreSQL and MySQL clusters across 54 data centers for 50+ Zoho applications, with multi-tenant provisioning for 50M+ tenants.
- Cut cluster failover from 1–2 minutes to roughly 5 seconds with a high-availability failover system in Go using witness-quorum consensus, removing manual split-brain checks during network partitions.
- Architected zero-downtime blue-green PostgreSQL upgrades, cutting spare capacity from 6 nodes per service to 2 and saving roughly 40 high-spec nodes per major upgrade.
- Engineered a distributed replication pipeline on PostgreSQL logical replication with a custom WAL decoder for cross-service DDL replay, eliminating schema-drift bugs the prior ORM-based decoder overlooked and running roughly 2.5x faster.
- Developed a cloud-scale backup and disaster recovery framework handling roughly 2 TB/week per cluster across PostgreSQL, MySQL, ClickHouse, etcd, Tarantool, and Prometheus.
- Optimized backend query routing by productionizing ProxySQL with TLS for SaaS clusters, reducing master load by roughly 40 percent on read-heavy applications.
- Resolved on-call production incidents across 900+ clusters in database (replication lag, deadlocks, query regressions, MVCC bloat) and network (TLS, VLAN, ProxySQL routing) layers via Prometheus, Grafana, and engine internals.
- Delivered zero-downtime VLAN migration for live production clusters by promoting a secondary as cutover target without breaking app connections.
- Leading a graph database POC evaluating DGraph, JanusGraph (Cassandra), and SurrealDB to give 50+ app teams native graph support inside our cluster topology.

Additional: PGConf India 2026, Bangalore Attendee
Networked with the ClickHouse co-founder; brought back PostgreSQL scaling patterns from 5+ companies. Reads database papers (Spanner, GFS, Neon).

PROJECTS

Cluster Management System Go, etcd
Built a backend orchestration service for MySQL and PostgreSQL clusters with witness-based quorum, automated failover, VIP migration, rolling upgrades, resumable workflows, and ProxySQL or PgBouncer integration; manages 900+ production clusters.

Internal-workings Go, C
Repo of systems primitives built from scratch: a paged **disk manager** in Go (bitmap free-space, file/mmap storage, page sync); a **linear-probing hash map** in Go; a 200-line C **memory allocator** (malloc/free with sbrk, mmap, free-lists); a C **network socket** server. github.com/LakshimiRam-073/Internal-workings

Personal Tech Blog stupidnotes.in
6+ long-form articles on database internals, system design, and low-level engineering.

KEY ACHIEVEMENTS

Google Foobar – Level 4

Reached Level 4 of Google's invite-only algorithmic challenge.

Smart India Hackathon

Finalist in the national-level innovation hackathon.

BIT Virupaksha

Finalist in the ML Hackathon at Bannari Amman Institute of Technology.

EDUCATION

B.E. Electronics and Communication Engineering
Sri Krishna College of Engineering and Technology Oct 2021 – May 2025
Coimbatore • CGPA 8.44

SKILLS

Languages — Go · Java · C · C++ · Python · SQL · Bash · Data Structures & Algorithms

Databases — PostgreSQL · MySQL · ClickHouse · Redis · etcd · Tarantool · DGraph · Query Optimization · Indexing · MVCC · WAL · Storage Engines (B+ Tree, LSM)

Distributed Systems & Backend — REST APIs · gRPC · Microservices · System Design · Consensus (Raft) · High Availability · Fault Tolerance · Concurrency · Data Consistency · Sharding · Logical Replication · Blue-Green Deployment

Cloud & Tools — AWS · Linux · Docker · Firecracker · ProxySQL · PgBouncer · Caching · Prometheus · Grafana · Observability · CI/CD · TLS/mTLS

INTERESTS

Database Internals

Reads research papers (Spanner, GFS, Neon) and writes long-form articles on storage and replication.

System Design

Studies large-scale architectures (sharding, consensus, replication trade-offs) and prototypes them in side projects.

Low-level Engineering

Building allocators, decoders, and tooling in C and Go from first principles.

LANGUAGES

English Professional • **Tamil** Native